

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458405.7139 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.046
Transit depth (Rp/Rs)^2	1.49 +/- 1.13 [%]
Orbital Inclination (inc)	85.83 +/- 2.95
Airmass coefficient 1 (a1)	0.95 +/- 0.61
Airmass coefficient 2 (a2)	0.05 +/- 0.2
Scatter in the residuals of the lightcurve fit is	69.27 %
Best Comparison Star	#3 - [504, 384]
Optimal Aperture	1.7
Optimal Annulus	7.0
Transit Duration (day)	0.071 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.122 ± 0.046 vs 0.1326 (deviation 0.15σ)
Duration fit vs published	0.071 d vs 0.0737 d (deviation 0.06σ)
Mid-time O-C	-12.6 min
Depth SNR	1.7
Residual scatter	69.27 %

Light curve

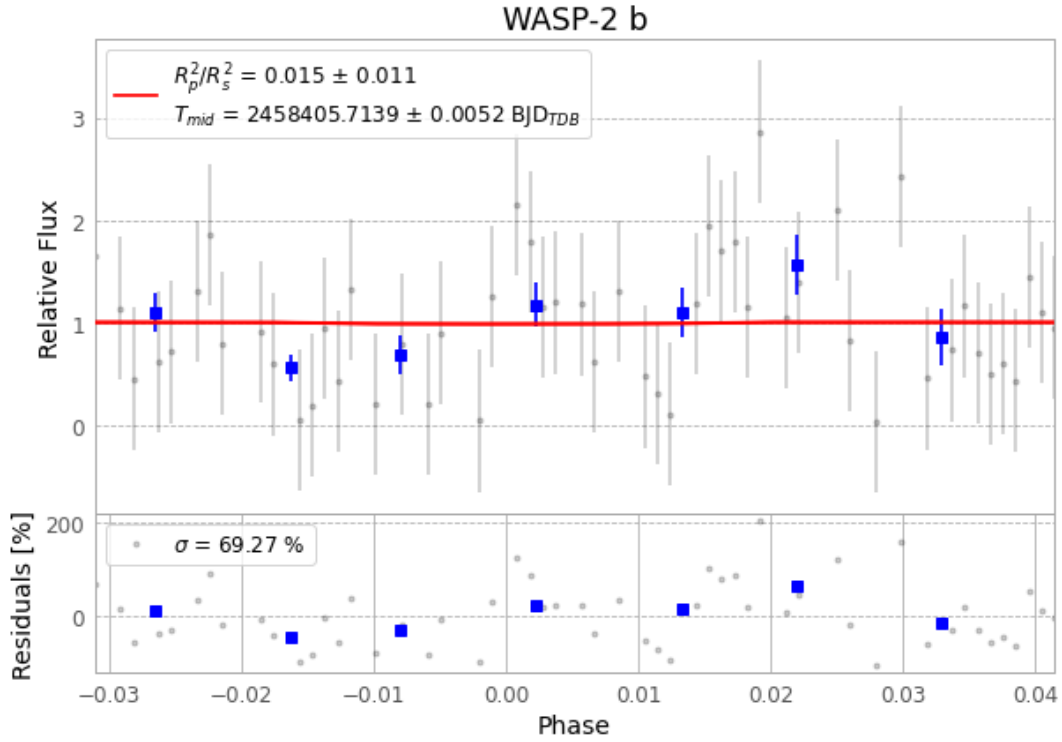


Figure 1 — FinalLightCurve_WASP-2 b_2018-10-13.png (SHA-256 de2083a5ffc4b1e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [565, 449], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 206e2417cadf45b4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

76 FITS frames (50 MB), WASP-2181014032712.FITS → WASP-2181014071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-181014.log (7833e739b514...), FinalLightCurve_WASP-2 b_2018-10-13.png (de2083a5ffc4...), FinalLightCurve_WASP-2 b_2018-10-13.csv (a62d9670900d...)

Compute resources

Wall time 240.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 21:26Z.